

Filter Concepts

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- The idea of filtering using LTI system is based on the convolution property of the DTFT which states that for LTI systems, the DTFT of the output $Y(e^{j\omega})$ is the product of the DTFT of the input $X(e^{j\omega})$ and the frequency response $H(e^{j\omega})$, i.e.,
$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega}).$$

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- In a sense, $H(e^{j\omega})$ acts as a weighting function or spectral shaping function to the different frequency components in the input signal.
- **Ideal Filters:** Ideal filters allow distortionless transmission of a certain band of frequencies and completely suppress the remaining frequencies. The passband of a filter is the band of frequencies that are passed by the system, while the stop band refers to the range of frequencies that are attenuated by the system. In the ideal case, $|H(e^{j\omega})| = 1$ in a passband, while $|H(e^{j\omega})| = 0$ in a stopband.

Low-pass, High-pass, Band-pass, Band-stop and All-pass Filters

- Lowpass filters are those characterized by a passband that extends from $\omega = 0$ to $\omega = \omega_c$, where ω_c is called the cutoff frequency of the filter [Fig. (a)]. A lowpass filter attenuates high frequency components of the input and passes the lower frequency components.

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- Highpass filters are characterized by a stopband that extends from $\omega = 0$ to $\omega = \omega_c$ and a passband that extends from $\omega = \omega_c$ to π [Fig. (b)]. A highpass filter attenuates low frequencies and passes the high frequencies.

Low-pass, High-pass, Band-pass, Band-stop and All-pass Filters

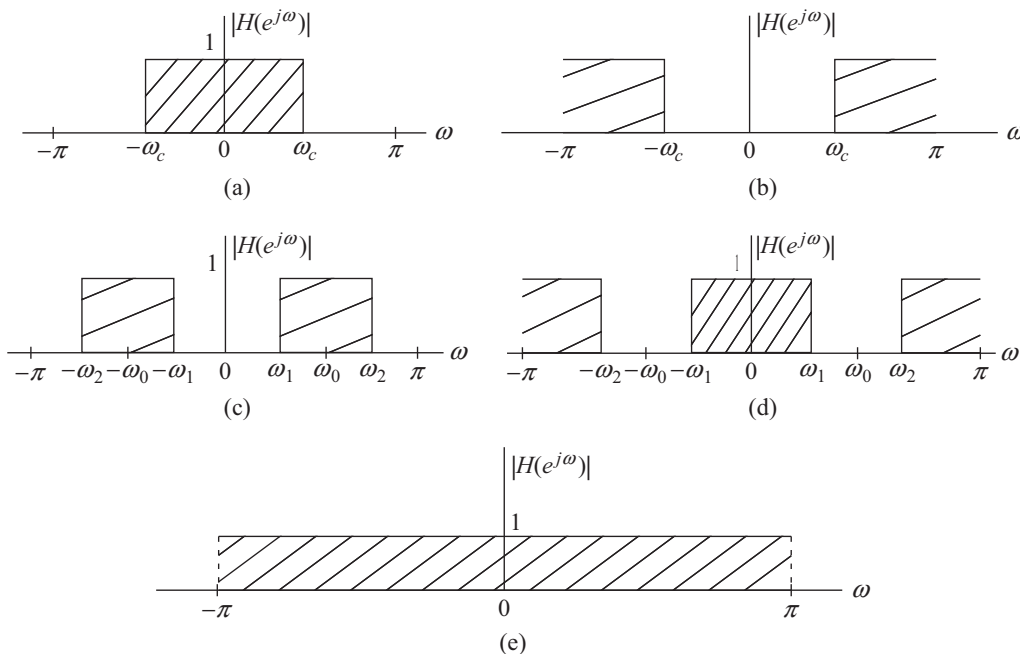
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- Bandpass filters are characterized by a passband that extends from $\omega = \omega_1$ to $\omega = \omega_2$, and all other frequencies are stopped [Fig. (c)]. A bandpass filter passes signals within a certain frequency band and attenuates signals outside that band.

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- Bandstop filters stop frequencies extending from $\omega = \omega_1$ to $\omega = \omega_2$ and pass all other frequencies [Fig. (d)]. A bandstop filter attenuates signals within a certain frequency band and passes signals outside that band.

Low-pass, High-pass, Band-pass, Band-stop and All-pass Filters

- An allpass filter is characterized by a magnitude response that is constant for all frequencies [Fig. (e)]. The characteristics of an allpass filter are completely determined by its phase-shift characteristics. An allpass filter has nonconstant group delay, different frequencies in the input are delayed by different amounts.



Linear-phase Filter

- **Magnitude and Phase response:** The frequency response of a digital filter is give by

$$H(e^{j\omega}) = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})}$$

where $|H(e^{j\omega})|$ is the magnitude response and $\angle H(e^{j\omega})$ is the phase response.

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1. Linear-phase filter
2. Zero-phase filter

- **Linear-Phase Filter:** Linear-phase filters (of length N) have either a *symmetric impulse response*, i.e.,

$$h(n) = h(N - 1 - n), \quad 0 \leq n \leq N - 1$$

or an *antisymmetric impulse response*, i.e.,

$$h(n) = -h(N - 1 - n), \quad 0 \leq n \leq N - 1$$

Zero-Phase Filter

● **Zero-Phase Filter:** A filter is said to be zero-phase when its frequency response $H(e^{j\omega})$ is a real and even function of frequency ω , and when $H(e^{j\omega}) > 0$ in the filter passband.

- A real frequency response has phase zero when it is positive, and phase π when it is negative. That is, if $H(e^{j\omega})$ is real, then

$$\angle H(e^{j\omega}) = \begin{cases} 0 & H(e^{j\omega}) > 0 \\ \pi & H(e^{j\omega}) < 0 \end{cases}$$

- A zero-phase filter cannot be causal (except in the trivial case when the filter is a constant scale factor $H(e^{j\omega}) = K$ or $h(n) = K\delta(n)$).
- However, in many non-real time applications, such as when filtering a sound file on a computer disk, causality is not a requirement, and zero-phase filters are often preferred.

Zero-Phase Filter

- The trivial filter $h(n) = \delta(n)$ has frequency response $H(e^{j\omega}) = 1$, which is zero phase for all ω .
- Every second-order zero-phase filter has an impulse response of the form

$$h(n) = b_1\delta(n+1) + b_0\delta(n) + b_1\delta(n-1)$$

where the coefficients b_i are assumed real. The transfer function of the general second-order real, zero-phase filter is

$$H(z) = b_1z + b_0 + b_1z^{-1}$$

and the frequency response is

$$H(e^{j\omega}) = b_1e^{j\omega} + b_0 + b_1e^{-j\omega} = b_0 + 2b_1\cos(\omega)$$

which is real for all ω .

Simple Lowpass FIR Digital Filter

● **Lowpass FIR Digital Filter:** The impulse response of a simple lowpass FIR filter is given by

$$h(n) = \frac{1}{2}[\delta(n) + \delta(n-1)]$$
$$H(z) = \frac{1}{2}(1 + z^{-1}) = \frac{z+1}{2z}$$

The system function has a zero at $z = -1$ and a pole at $z = 0$. The frequency response of the filter is given by

$$H(e^{j\omega}) = \frac{1}{2}(1 + e^{-j\omega}) = e^{-j\omega/2} \left(\frac{e^{j\omega/2} + e^{-j\omega/2}}{2} \right)$$
$$H(e^{j\omega}) = e^{-j\omega/2} \cos\left(\frac{\omega}{2}\right) = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})} \quad 0 \leq \omega \leq \pi$$

The magnitude response is given by

$$|H(e^{j\omega})| = \cos\left(\frac{\omega}{2}\right) \quad 0 \leq \omega \leq \pi$$

Simple Lowpass FIR Digital Filter contd...

- The magnitude response $|H(e^{j\omega})|$ starts from 1 at $\omega = 0$ and monotonically decreasing to zero at $\omega = \pi$.

$$|H(e^{j\omega})| = \begin{cases} 1 & \omega = 0 \\ 0 & \omega = \pi \end{cases}$$

- The frequency $\omega = \omega_c$ at which $|H(e^{j\omega_c})| = \frac{1}{\sqrt{2}}$ is known as cut-off frequency (or 3-dB frequency). That is

$$|H(e^{j\omega_c})| = \cos\left(\frac{\omega_c}{2}\right) = \frac{1}{\sqrt{2}} = \cos\left(\frac{\pi}{4}\right)$$

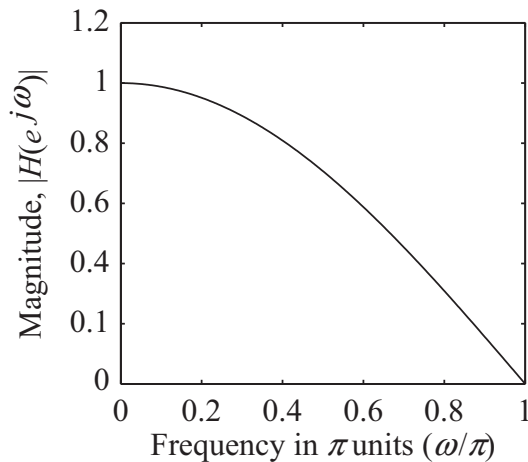
which yields

$$\frac{\omega_c}{2} = \frac{\pi}{4} \quad \text{or} \quad \omega_c = \frac{\pi}{2}$$

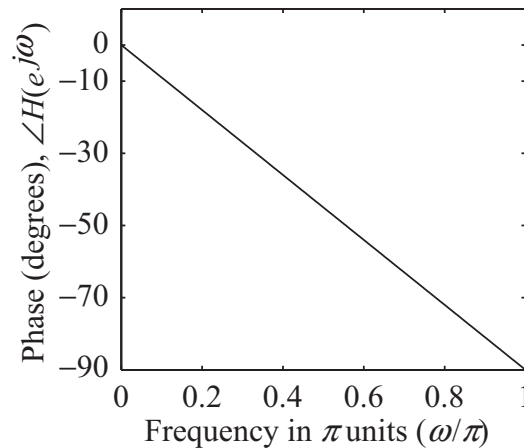
Simple Lowpass FIR Digital Filter contd...

- The 3-dB cutoff frequency ω_c can be considered as the passband edge frequency. For this filter the passband is from 0 to $\frac{\pi}{2}$ and stopband is from $\frac{\pi}{2}$ to π .
- The phase response is given by

$$\angle H(e^{j\omega}) = -\frac{\omega}{2} \quad 0 \leq \omega \leq \pi$$



(a)



(b)

Simple Highpass FIR Digital Filter

● **Highpass FIR Digital Filter:** The impulse response of a simple highpass FIR filter is given by

$$h(n) = \frac{1}{2}[\delta(n) - \delta(n-1)]$$
$$H(z) = \frac{1}{2}(1 - z^{-1}) = \frac{z-1}{2z}$$

The system function has a zero at $z = 1$ and a pole at $z = 0$. The frequency response of the filter is given by

$$H(e^{j\omega}) = \frac{1}{2}(1 - e^{-j\omega}) = je^{-j\omega/2} \left(\frac{e^{j\omega/2} - e^{-j\omega/2}}{2j} \right)$$
$$H(e^{j\omega}) = je^{-j\omega/2} \sin\left(\frac{\omega}{2}\right) = e^{j(\frac{\pi}{2} - \frac{\omega}{2})} \sin\left(\frac{\omega}{2}\right) = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})} \quad 0 \leq \omega \leq \pi$$

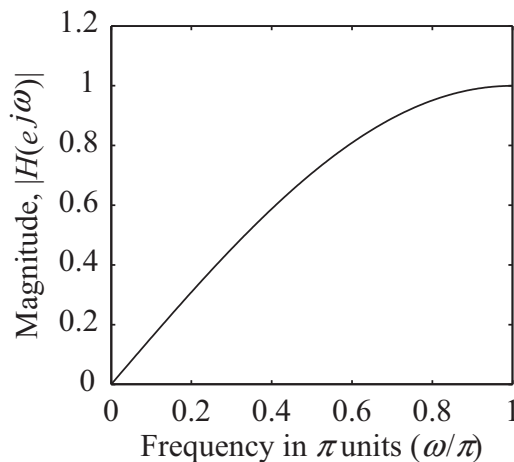
The magnitude response is given by

$$|H(e^{j\omega})| = \sin\left(\frac{\omega}{2}\right) \quad 0 \leq \omega \leq \pi$$

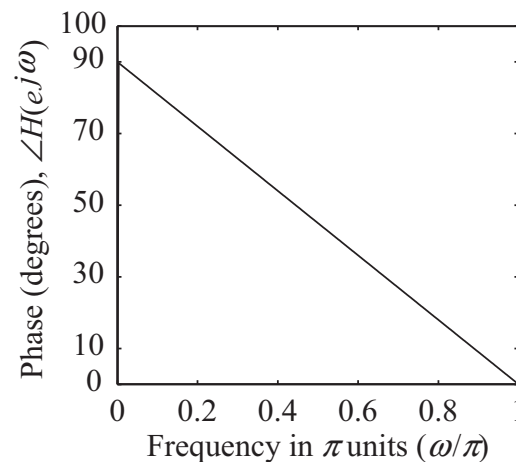
Simple Highpass FIR Digital Filter contd...

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- The phase response is given by

$$\angle H(e^{j\omega}) = \frac{\pi}{2} - \frac{\omega}{2} \quad 0 \leq \omega \leq \pi$$



(a)



(b)

Simple Bandpass FIR Digital Filter

● **Bandpass FIR Digital Filter:** The impulse response of a simple lowpass FIR filter is given by

$$h(n) = \frac{1}{2}[\delta(n) - \delta(n - 2)]$$
$$H(z) = \frac{1}{2}(1 - z^{-2}) = \frac{z^2 - 1}{2z^2}$$

The system function has zeros at $z = \pm 1$ and double poles at $z = 0$. The frequency response of the filter is given by

$$H(e^{j\omega}) = \frac{1}{2}(1 - e^{-j2\omega}) = je^{-j\omega} \left(\frac{e^{j\omega} - e^{-j\omega}}{2j} \right)$$
$$H(e^{j\omega}) = je^{-j\omega} \sin(\omega) = e^{j\pi/2} e^{-j\omega} \sin(\omega)$$
$$H(e^{j\omega}) = e^{j(\frac{\pi}{2} - \omega)} \sin(\omega) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})} \quad 0 \leq \omega \leq \pi$$

The magnitude response is given by

$$|H(e^{j\omega})| = \sin(\omega) \quad 0 \leq \omega \leq \pi$$

Simple Bandpass FIR Digital Filter contd...

- A bandpass filter has two cutoff frequencies: lower cutoff frequency (ω_l) and upper cutoff frequency (ω_u) and $\omega_l < \omega_u$. The difference, $\Delta\omega = \omega_u - \omega_l$, is known as the bandwidth of the bandpass filter. Thus

$$|H(e^{j\omega})| \Big|_{\omega=\omega_l, \omega_u} = \sin(\omega) \Big|_{\omega=\omega_l, \omega_u} = \frac{1}{\sqrt{2}} = \sin\left(\frac{\pi}{4}\right) = \sin\left(\frac{3\pi}{4}\right)$$

which yields

$$\omega_l = \frac{\pi}{4} \quad \text{and} \quad \omega_u = \frac{3\pi}{4}$$

and the bandwidth is given by

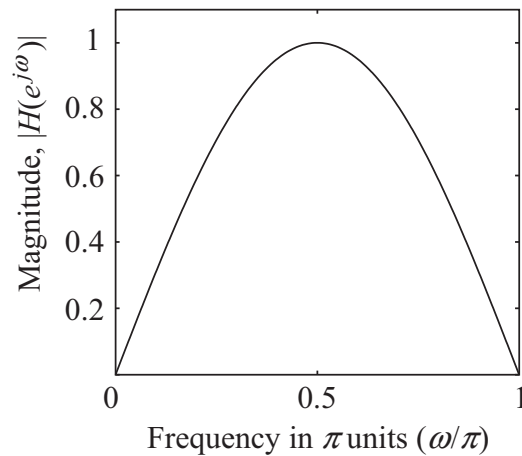
$$\Delta\omega = \omega_u - \omega_l = \frac{3\pi}{4} - \frac{\pi}{4} = \frac{\pi}{2}$$

The bandpass filter has two stopbands: one is from 0 to $\frac{\pi}{4}$ and other is from $\frac{3\pi}{4}$ to π , and one passband which is from $\frac{\pi}{4}$ to $\frac{3\pi}{4}$.

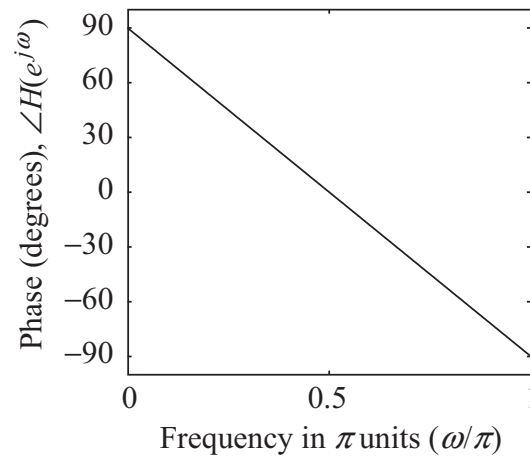
Simple Bandpass FIR Digital Filter contd...

- The phase response is given by

$$\angle H(e^{j\omega}) = \frac{\pi}{2} - \omega \quad 0 \leq \omega \leq \pi$$



(a)



(b)

Simple Bandstop FIR Digital Filter

● **Bandstop FIR Digital Filter:** The impulse response of a simple lowpass FIR filter is given by

$$h(n) = \frac{1}{2}[\delta(n) + \delta(n - 2)]$$
$$H(z) = \frac{1}{2}(1 + z^{-2}) = \frac{z^2 + 1}{2z^2}$$

The system function has zeros at $z = \pm j$ and double poles at $z = 0$. The frequency response of the filter is given by

$$H(e^{j\omega}) = \frac{1}{2}(1 + e^{-j2\omega}) = e^{-j\omega} \left(\frac{e^{j\omega} + e^{-j\omega}}{2} \right)$$
$$H(e^{j\omega}) = e^{-j\omega} \cos(\omega) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})} \quad 0 \leq \omega \leq \pi$$

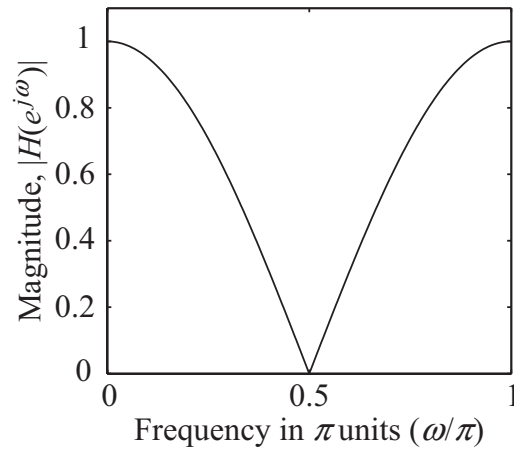
The magnitude response is given by

$$|\cos(\omega)| \quad 0 \leq \omega \leq \pi$$

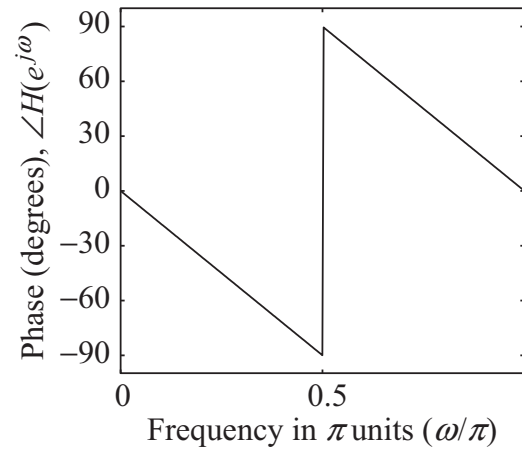
Simple Bandstop FIR Digital Filter contd...

- The phase response is given by

$$\angle H(e^{j\omega}) = \begin{cases} -\omega & 0 \leq \omega \leq \frac{\pi}{2} \\ -\omega + \pi & \frac{\pi}{2} < \omega \leq \pi \end{cases}$$



(a)



(b)

Simple Lowpass IIR Digital Filter

- **Lowpass IIR Digital Filter:** The system function of a first order lowpass IIR digital filter is given by

$$H(z) = \frac{1-a}{2} \frac{1+z^{-1}}{1-az^{-1}}$$

where $|a| < 1$ for stability. The system function has a zero at $z = -1$ and a pole at $z = a$.

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- The frequency response is given by

$$H(e^{j\omega}) = \frac{1-a}{2} \frac{1+e^{-j\omega}}{1-ae^{-j\omega}} = \frac{1-a}{2} \left(\frac{1+\cos\omega - j\sin\omega}{1-a\cos\omega + ja\sin\omega} \right)$$

where

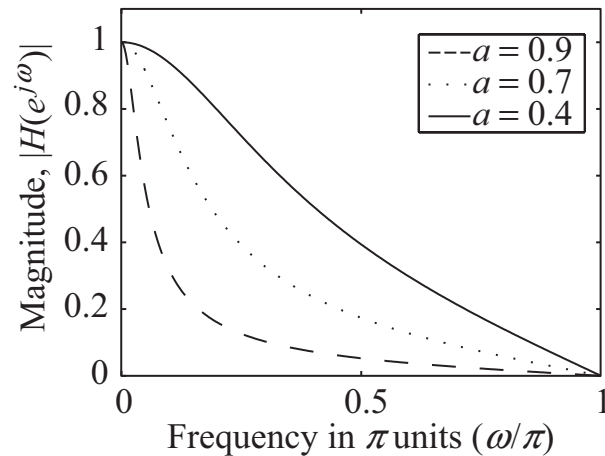
$$a = \frac{1 - \sin\omega_c}{\cos\omega_c}$$

where ω_c is the 3-dB cutoff frequency.

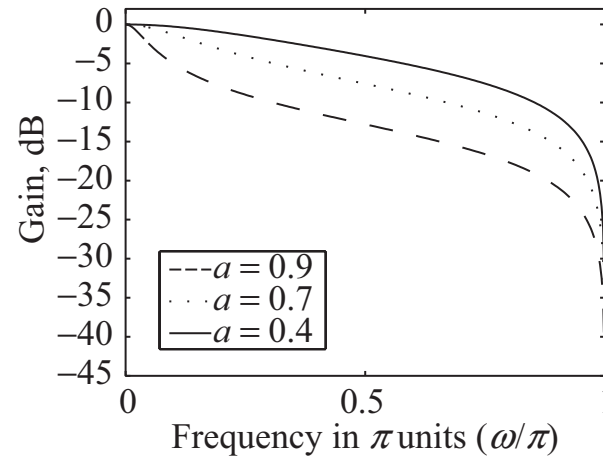
Simple Lowpass IIR Digital Filter contd...

● The magnitude response is given by

$$|H(e^{j\omega})| = \frac{1-a}{2} \left(\frac{[2 + 2\cos\omega]^{1/2}}{[1 + a^2 - 2a\cos\omega]^{1/2}} \right)$$



(a)



(b)

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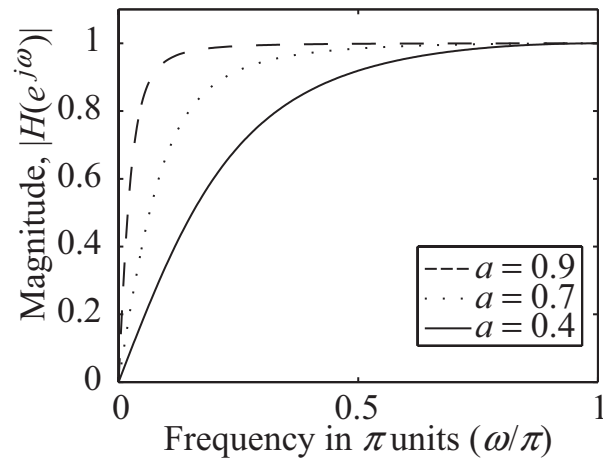
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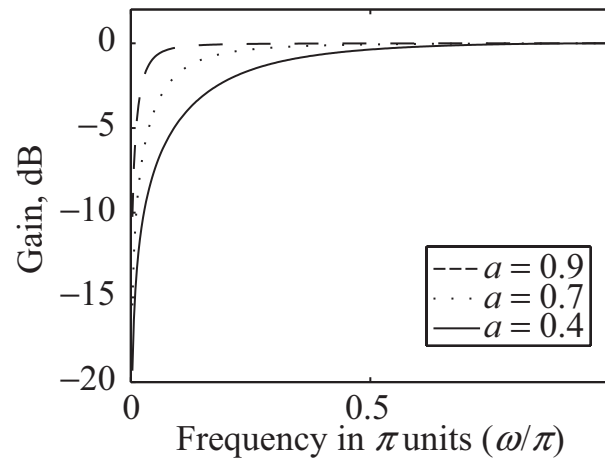
Simple Highpass IIR Digital Filter contd...

- The magnitude response is given by

$$|H(e^{j\omega})| = \frac{1+a}{2} \left(\frac{[2 - 2\cos\omega]^{1/2}}{[1 + a^2 - 2a\cos\omega]^{1/2}} \right)$$



(a)



(b)

Simple Bandpass IIR Digital Filter

● **Bandpass IIR Digital Filter:** The system function of a second order bandpass IIR digital filter is given by

$$H(z) = \frac{1-a}{2} \frac{1-z^{-2}}{1-b(1+a)z^{-1}+az^{-2}}$$

The system function has two zeros located at $z = \pm 1$ which indicates that the dc response and high frequency response are zero.

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- The frequency response is given by

$$H(e^{j\omega}) = \frac{1-a}{2} \frac{1-e^{-j2\omega}}{1-b(1+a)e^{-j\omega}+ae^{-j2\omega}}$$

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- The magnitude response $|H(e^{j\omega})|$ is given by

$$|H(e^{j\omega})| = \frac{1-a}{2} \left(\frac{[(1-\cos 2\omega)^2 + \sin^2 2\omega]^{1/2}}{[(1-b(1+a)\cos \omega + a\cos 2\omega)^2 + (b(1+a)\sin \omega - a\sin 2\omega)^2]^{1/2}} \right)$$

Simple Bandpass IIR Digital Filter contd...

- The center frequency ω_0 where the magnitude value is unity, is given by

$$\omega_0 = \cos^{-1}(b)$$

Simple Bandpass IIR Digital Filter contd...

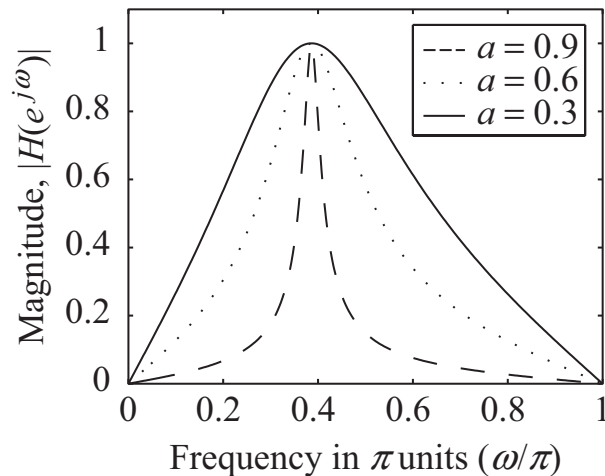
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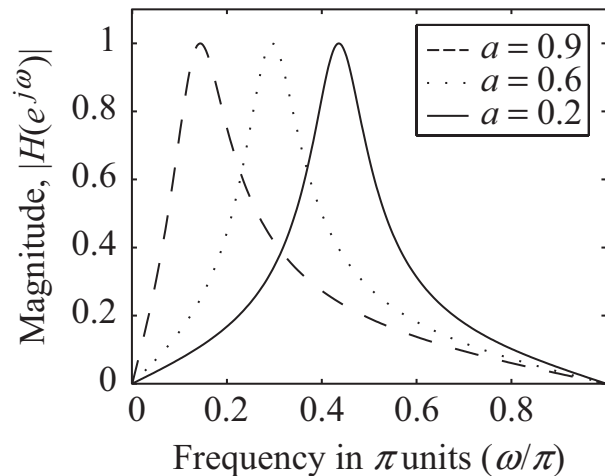
- The bandwidth of the filter which is equal to the difference of two cutoff frequencies ω_u and ω_l ($\omega_u > \omega_l$) is given by

$$\Delta\omega = \omega_u - \omega_l = \cos^{-1} \left(\frac{2a}{1 + a^2} \right)$$

Note that the two constants b & a are independent of each other.



(a)



(b)

Simple Bandstop IIR Digital Filter

- **Bandstop IIR Digital Filter:** The system function of a second order bandstop IIR digital filter is given by

$$H(z) = \frac{1+a}{2} \frac{1 - 2bz^{-1} + z^{-2}}{1 - b(1+a)z^{-1} + az^{-2}}$$

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- The frequency response of the filter is given by

$$H(e^{j\omega}) = \frac{1+a}{2} \frac{1 - 2be^{-j\omega} + e^{-j2\omega}}{1 - b(1+a)e^{-j\omega} + ae^{-j2\omega}}$$

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- The frequency response of the filter is given by

$$H(e^{j\omega}) = \frac{1+a}{2} \frac{1 - 2be^{-j\omega} + e^{-j2\omega}}{1 - b(1+a)e^{-j\omega} + ae^{-j2\omega}}$$

- Note that $H(1) = 1$ and also $H(-1) = 1$. That is, the magnitude response takes the maximum value of unity at $\omega = 0$ and $\omega = \pi$.

Simple Bandstop IIR Digital Filter contd...

- At notch frequency ω_0 , the magnitude response goes to zero, which is given by

$$\omega_0 = \cos^{-1}(b)$$

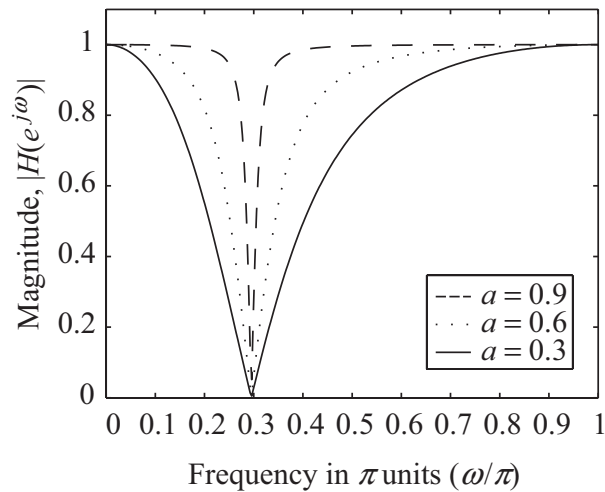
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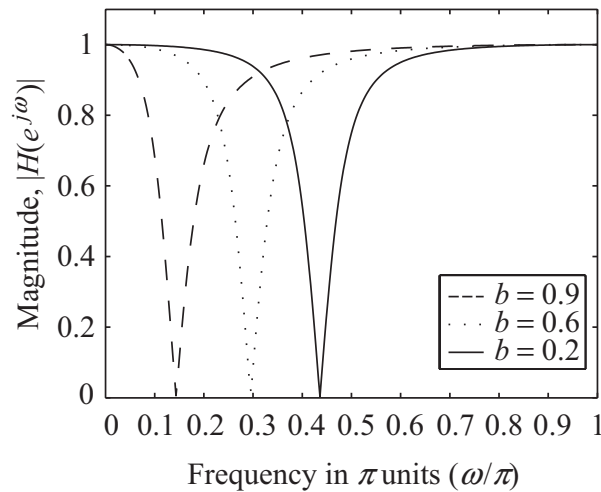
$$\omega_0 = \cos^{-1}(b)$$

- The bandwidth of the filter which is equal to the difference of two cutoff frequencies ω_u and ω_l ($\omega_u > \omega_l$) is given by

$$\Delta\omega = \omega_u - \omega_l = \cos^{-1} \left(\frac{2a}{1 + a^2} \right)$$



(a)



(b)

Allpass Filter

- **Allpass Filter:** An *allpass filter* is a filter that passes all frequency components equally because it has a constant magnitude response for all frequencies, that is,

$$|H_{\text{ap}}(e^{j\omega})| = 1 \quad 0 \leq \omega \leq \pi$$

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- An N th order causal real-coefficient allpass filter is described by the system function

$$H_{\text{ap}}(z) = \frac{a_N + a_{N-1}z^{-1} + \dots + a_1z^{-N+1} + z^{-N}}{1 + a_1z^{-1} + \dots + a_{N-1}z^{-N+1} + a_Nz^{-N}}$$

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- An allpass filter of order N has a numerator and denominator of equal order N with coefficients in reverse order. If we define the polynomial $A(z)$ as

$$A(z) = 1 + a_1z^{-1} + \dots + a_{N-1}z^{-N+1} + a_Nz^{-N}$$

then it follows that

$$z^{-N}A(z^{-1}) = a_N + a_{N-1}z^{-1} + \dots + a_1z^{-N+1} + z^{-N}$$

Allpass Filter contd...

 Hence

$$H_{\text{ap}}(z) = \frac{z^{-N} A(z^{-1})}{A(z)}$$

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● Consider a filter $H(z)$ with a desired magnitude response but an unsatisfactory phase response. In this case, we may connect it to an allpass filter $H_{\text{ap}}(z)$. Then we have

$$|H(e^{j\omega})H_{\text{ap}}(e^{j\omega})| = |H(e^{j\omega})||H_{\text{ap}}(e^{j\omega})| = |H(e^{j\omega})|$$

and

$$\angle(H(e^{j\omega})H_{\text{ap}}(e^{j\omega})) = \angle(H(e^{j\omega})) + \angle H_{\text{ap}}(e^{j\omega})$$

Because the magnitude response of $H(z)$ is not affected by introducing an allpass filter, we can use an allpass filter to improve the phase response of $H(z)H_{\text{ap}}(z)$.

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2. A causal stable allpass filter with N poles has a phase response $\angle H_{\text{ap}}(e^{j\omega})$ that decreases monotonically from $[\angle H_{\text{ap}}(e^{j0})]$ to $[\angle H_{\text{ap}}(e^{j0}) - 2\pi N]$ as ω increases from 0 to 2π .

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3. A stable allpass filters have nonnegative group delay, that is, $\tau_g(\omega) \geq 0$.
4. Since a stable allpass filter has all its poles inside the unit circle, all its zeros outside, and exhibits unit magnitude along the unit circle, one can deduce

$$|H_{\text{ap}}(z)| \begin{cases} < 1 & \text{if } |z| > 1 \\ = 1 & \text{if } |z| = 1 \\ > 1 & \text{if } |z| < 1 \end{cases}$$

● **Minimum-Phase System:** An IIR system with system function $H(z)$ is called *minimum-phase* if all its poles and zeros are inside the unit circle. The minimum-phase system shows the smallest group delay at every frequency among all systems with the same magnitude response.

Minimum-Phase System	$H_1(z) = \frac{(z - \frac{1}{2})(z - \frac{1}{4})}{(z - \frac{1}{3})(z - \frac{1}{5})}$
Maximum-Phase System	$H_2(z) = \frac{(1 - \frac{1}{2}z)(z - \frac{1}{4})}{(z - \frac{1}{3})(z - \frac{1}{5})}$
Mixed-Phase System	$H_3(z) = \frac{(1 - \frac{1}{2}z)(1 - \frac{1}{4}z)}{(z - \frac{1}{3})(z - \frac{1}{5})}$

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- **Mixed-Phase System:** A system is called the *mixed-phase* if some, but not all, of the zeros are outside the unit circle.

Cepstrum and Homomorphic Deconvolution

● **Cepstrum:** Consider a signal $x(n)$ having DTFT $X(e^{j\omega})$. The *complex cepstrum* (note that cepstrum is “spectrum” with the first syllable flipped) of the signal $x(n)$ is defined as the signal $c_x(n)$, which is the inverse DTFT of $C_x(e^{j\omega})$, where

$$C_x(e^{j\omega}) = \ln X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} c_x(n) e^{-j\omega n}$$

where

$$c_x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [\ln X(e^{j\omega})] e^{j\omega n} d\omega \quad (1)$$

If we express $X(e^{j\omega})$ in terms of its magnitude and phase, say

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\theta(\omega)}$$

then

$$C_x(e^{j\omega}) = \ln X(e^{j\omega}) = \ln |X(e^{j\omega})| + j\theta(\omega)$$

Cepstrum contd...

● The cepstrum $c_x(n)$ is expressed as

$$\begin{aligned}c_x(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} [\ln |X(e^{j\omega})| + j\theta(\omega)] e^{j\omega n} d\omega \\&= \frac{1}{2\pi} \int_{-\pi}^{\pi} \ln |X(e^{j\omega})| e^{j\omega n} d\omega + j \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta(\omega) e^{j\omega n} d\omega \\c_x(n) &= c_m(n) + jc_\theta(n)\end{aligned}$$

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- The cepstrum $c_x(n)$ is a complex quantity. Its real part $c_m(n)$ corresponds to the IDTFT of $\ln |X(e^{j\omega})|$ and is called the *real cepstrum*.
- In some applications, such as speech processing, only the component $c_m(n)$ is computed. In speech processing the real cepstrum $c_m(n)$ has been used to separate and thus to estimate the spectral content of the speech from the pitch frequency of the speech.

Homomorphic Deconvolution

- The complex cepstrum is used in practice to separate signals that are convolved. The process of separating two convolved signals is called *deconvolution* and the use of complex cepstrum to perform the separation is called *homomorphic deconvolution*.

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- Let $x(n)$ and $y(n)$ be the input and output signal of an LTI system. Then

$$y(n) = x(n) * h(n), \quad \text{or} \quad Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

where $H(e^{j\omega})$ is the frequency response. The logarithm of $Y(e^{j\omega})$ is

$$C_y(e^{j\omega}) = \ln Y(e^{j\omega}) = \ln X(e^{j\omega}) + \ln H(e^{j\omega}) = C_x(e^{j\omega}) + C_h(e^{j\omega})$$

Inverse transformation gives

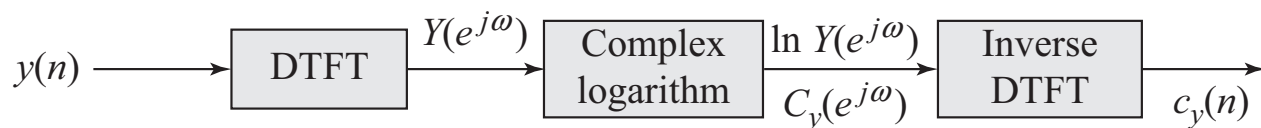
$$c_y(n) = c_x(n) + c_h(n)$$

Homomorphic Deconvolution contd...

- The convolution operation transforms to a summation in the cepstral domain.

$$y(n) = x(n) * h(n) \qquad c_y(n) = c_x(n) + c_h(n)$$

The system performing these transformations is called homomorphic system and is illustrated in Figure.

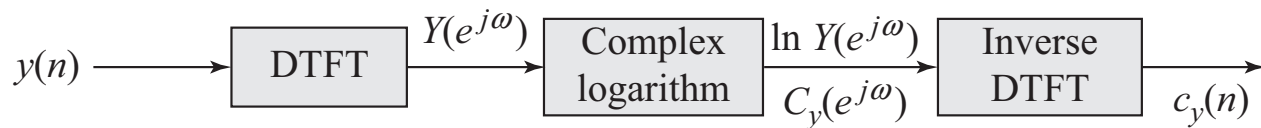


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- Cepstral methods and homomorphic filtering have found practical applications in various fields including deconvolution, image processing (restoring degraded images), communications (echo cancelation), speech processing and seismic signal processing.

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- Consider a causal N -point averaging filter whose difference equation is given by

$$y(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(n-k) = \frac{1}{N} [x(0) + x(1) + \cdots x(n-N+1)]$$

Its impulse response is given by

$$h(n) = \begin{cases} \frac{1}{N} & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

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- The system function of the filter is given by

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k} = \frac{1}{N} \left[\frac{1 - z^{-N}}{1 - z^{-1}} \right] = \frac{1}{N} \left[\frac{z^N - 1}{z^{N-1}(z - 1)} \right]$$

Averaging Filter contd...

- The frequency response is given by

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}} = \frac{1}{N} \left(\frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}} \right) = \frac{1}{N} e^{-j\omega(N-1)/2} \left(\frac{\sin \frac{\omega N}{2}}{\sin \frac{\omega}{2}} \right)$$

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- The filter has zeros on the unit circle at

$$z^N - 1 = 0$$

$$z^N = 1 = e^{j\frac{2\pi}{N}kN}, \quad k = 1, 2, 3, \dots, N-1$$

$$z = e^{j\frac{2\pi}{N}k}, \quad k = 1, 2, 3, \dots, N-1$$

The N roots of $z^N - 1 = 0$ lie on the unit circle with an angular spacing of $\frac{2\pi}{N}$ radians starting at $\omega = 0$ (corresponding to $z = 1$). However, the zero at $z = 1$ is canceled by the pole at $z = 1$. So, we have $N - 1$ zeros on the unit circle and $N - 1$ poles at the origin ($z = 0$).

Comb Filters

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Comb Filters

- **Comb Filter:** A filter which has multiple passbands and stopbands is known as comb filter. A comb filter has a magnitude response that looks much like the rounded teeth of a comb.
- A comb filter has a frequency response that is a periodic function of ω with a period $\frac{2\pi}{L}$, where L is a positive integer.
- A comb filter can be generated from either an FIR or an IIR filter. If $H(z)$ is a filter with a single passband and/or a single stopband, a comb filter can easily be generated from it by replacing z with z^L . The system function (and frequency response) of the comb filter is given by

$$G(z) = H(z^L), \quad \text{and} \quad G(e^{j\omega}) = H(e^{j\omega L})$$

Consequently, the frequency response characteristic $G(e^{j\omega})$ is simply an L -order repetition of $H(e^{j\omega})$ in the range $0 \leq \omega \leq 2\pi$.

Comb Filters contd...

- For example, consider a lowpass FIR filter.

$$H_0(z) = \frac{1}{2}(1 + z^{-1})$$

The comb filter generated from $H_0(z)$ has a transfer function

$$G_0(z) = H_0(z^L) = \frac{1}{2}(1 + z^{-L})$$

Its frequency response is given by

$$G_0(e^{j\omega}) = \frac{1}{2}(1 + e^{-j\omega L}) = e^{-j\omega L/2} \cos \frac{\omega L}{2}$$

$$|G_0(e^{j\omega})| = \left| \cos \frac{\omega L}{2} \right|$$

Comb Filters contd...

- The comb filter $G_1(z)$ generated from a highpass FIR filter which has a transfer function

$$H_1(z) = \frac{1}{2}(1 - z^{-1})$$

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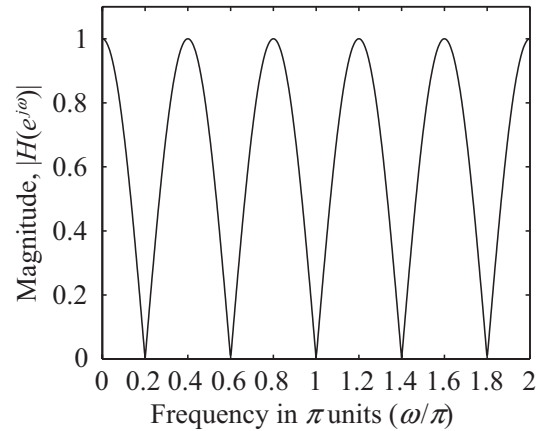
Its frequency response is given by

$$G_1(e^{j\omega}) = \frac{1}{2}(1 - e^{-j\omega L}) = je^{-j\omega L/2} \sin \frac{\omega L}{2}$$

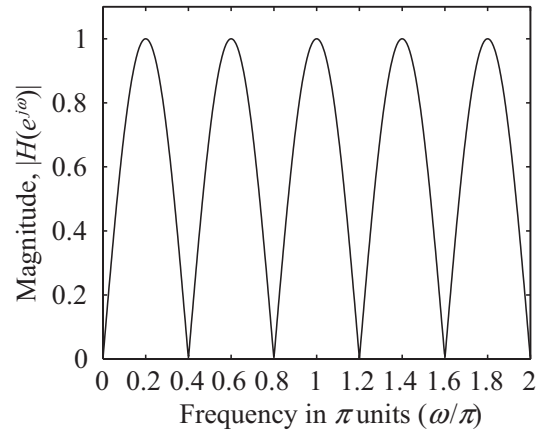
$$|G_1(e^{j\omega})| = \left| \sin \frac{\omega L}{2} \right|$$

Comb Filters contd...

- Magnitude response of FIR comb filters is shown in Figure generated from a lowpass filter with $L = 5$, and (b) generated from a highpass filter of with $L = 5$.



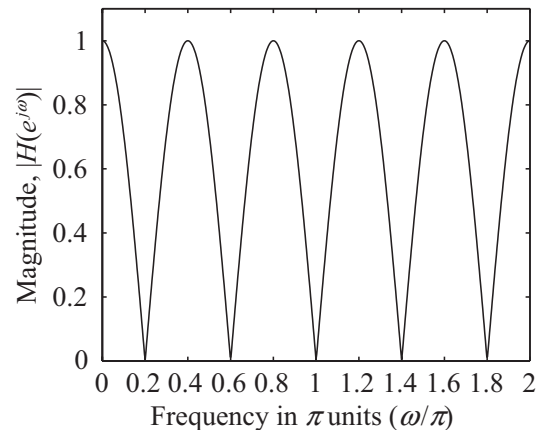
(a)



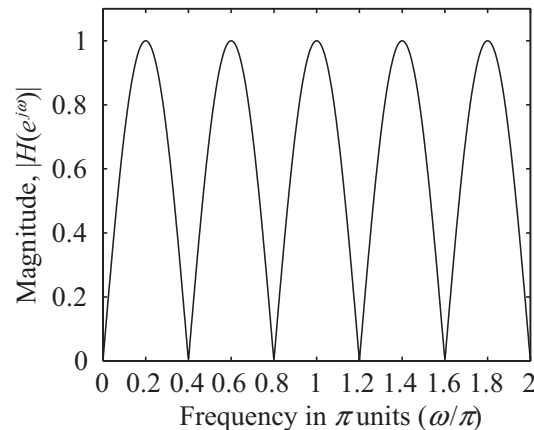
(b)

Comb Filters contd...

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(a)



(b)

- Applications:
 - (i) in creating special audio effects in musical sound processing,
 - (ii) in the rejection of power-line harmonics,
 - (iii) in the suppression of clutter from fixed objects in moving-target-indicator (MTI) radars.
 - (iv) Comb filter with multiple notch frequencies find applications in the cancelation of periodic interferences.

Filter Concepts

Thank You!